Name	Mohammed Rehan Moyal	Session	Sem - I
Subject	C - LAB	Roll No.	
Class	BCA - 1st - G I	Semester	1st Sem.

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→ Assignment ⇒ 01 ⇒ Write a program to print a simple message.

→ Code :-

```
#Include <stdio.h>
    int main()
        { printf("Hello, World");
          return 0;
        }
```

→ Output :- Hello World.

→ Assignment ⇒ 02 ⇒ Write a program to display your name, Roll no., Course & College Name.

→ Code :-

```
#Include <stdio.h>
    int main()
        { printf(" My Data \n");
          print f (" Name :- Rehan \n");
          printf (" Roll no. :- 21 \n");
          printf (" Course :- BCA \n");
          print f (" College :- Lachoo \n");
          return 0;
        }
```

→ Output :-

My Data
Name :- Rehan
Roll no :- 21
Course :- BCA
College :- Lachoo

→ Assignment ⇒ 03 ⇒ Write a program to swap two integers Using temp. variable.

→ Code ⇒

```
#include <stdio.h>
    int main()
    { int a, b, c;
      printf("Enter value of A :");
      scanf("%d", &a);
      printf("Enter value of B :");
      scanf("%d", &b);
      c = a;
      a = b;
      b = c;
      printf("Value of A = %d \n Value of B = %d \n
              , a, b);
      return 0;
    }
```

→ Output :-

Enter value of A :- 2
Enter value of B :- 5
Value of A :- 5
Value of B :- 2

→ Assignment ⇒ 04 ⇒ Write a program to Calc. Area & Circumference of Circle.
Take Radius input from user.

→ Code :-

```
#include <stdio.h>
int main()
{ int r;
  float a, c;
  printf("Enter Radius :- ");
  scanf("%d", &r);
  a = (3.141) * r * r;
  c = 2 * (3.141) * r;
  printf("Area = %.2f, \n Circumference = %.2f, "
         , a, c);
  return 0; }
```

→ Output :- Enter the radius :- 5
Area = 78.53
Circumference = 31.41.

→ Assignment ⇒ 05 ⇒ Write a program to calculate Area and Perimeter of Rectangle. Taking Length and Breadth input from user.

→ Code :-

```
# include <stdio.h>
    int main()
      { float l, b, a, p;
        printf("Enter the Length :- ");
        scanf("%f", &l);
        printf("Enter the Breadth :-");
        scanf("%f", &b);
        a = l * b;
        p = 2*(l+b);
        printf("Area = %.2f; \n Perimeter = %.2f;"
               , a, p);
      return 0;
      }
```

→ Output :- Enter the Length :- 5
Enter the Breadth :- 4
Area = 20.00
Perimeter = 18.00

→ Assignment ⇒ 06 ⇒ Write a program to Convert Temperature from Celsius to Fahrenheit and vice-versa. Take respective Temp. input from user.

→ Code :-

```
#include <stdio.h>
int main()
{ int A;
  float c, f;
  printf("Press 1 for C to F \n Press 2 for F to C :- ");
  scanf("%d", &A);

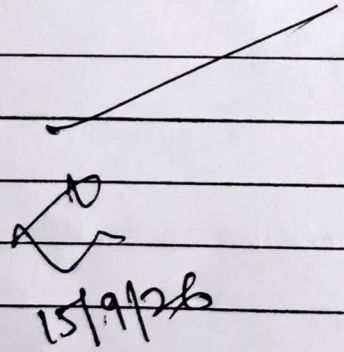
  if (A == 1)
   { printf("Celsius :- ");
     scanf("%f", &c);
     f = (c * 1.8) + 32;
     printf("%.2f degree Celsius is %.2f in Fahrenheit",
            c, f);
   }
  if (A == 2)
   { printf("Fahrenheit :- ");
     scanf("%f", &f);
     c = (f - 32) * (0.55555);
     printf("%.2f degree in Fahrenheit is %.2f in
            Celsius", f, c);
   }
  return 0;
}
```

→ Output :- I :-

Press 1 for C to F,
Press 2 for F to C :- 1
Celsius :- 100
Fahrenheit :- 212.00

→ Output :- II :-

Press 1 for C to F,
Press 2 for F to C :- 2
Fahrenheit :- 212
Celsius :- 100.00



15/9/26

→ Assignment ⇒ 07 ⇒ Write a program to find the largest of three numbers.

→ Code :-

```
#include <stdio.h>
int main()
{int n1, n2, n3;
 printf("Enter 1st :-");
 scanf("%d", &n1);
 printf("Enter 2nd :- ");
 scanf("%d", &n2);
 printf("Enter 3rd :- ");
 scanf("%d", &n3);

if(n1>n2)
 { if(n1>n3)
   { printf("1st is the largest.");
   }}

if(n2>n3)
  {if(n2>n3)
    { printf("2nd is the largest.");
    }}

if(n3>n1)
  {if(n3>n2)
    { printf("3rd is the largest.");
    }}
 return 0;
}
```

→ Output:- Enter 1st :- 2
Enter 2nd :- 3
Enter 3rd :- 1
2nd is the Largest.

→ Assignment → 08 → Write a program to input a student's mark and assign grades (A, B, C, D, F). use if else-if ladder.

→ Code →

```
#Include <stdio.h>
int main()
{ int x;
  printf("Assigned Marks :- ");
  scanf("%d",&x);

 if (x<=59 && x>=0)
  {printf("you Got an F.");}
 if (x<=69 && x>=60)
  {printf("you Got a D.");}
 if (x<=79 && x>=70)
  {printf("you Got a C.");}
 if(x<=89 && x>=80)
  {printf("you Got a B");}
 if(x<=100 && x>=90)
  {printf("you Got an A");}
 return 0;
}
```

→ Output :- Assigned Marks :- 89
yah Got a B.

→ Assignment → 09 → Write a program to demonstrate Switch-Case Statement.

→ Code :-

```
# include <stdio.h>
    int main()
    { int x;
      printf("Assigned Marks :- ");
      scanf("%d", &x)

      switch (x)
      { case 0 ... 49 ;
        printf("you failed");
        break;

        case 50 ... 100;
        printf("you Passed");
        break;
      }
      return 0;
    }
```

→ Output :- Assigned Marks :- 52
you Passed.

→ Assignment ⇒ 10 ⇒ Write a program to check if a given no. is Odd or Even.

→ Code ⇒

```
# include <stdio.h>
   int main()
    { int n;
      printf("Enter a no.:- ");
      Scanf("%d",&n)

    if (n%2==0)
     { printf(" No. is Even.");}
    else
     { printf(" No. is Odd.");}
    return 0;
   }
```

→ Output ⇒ Enter a No.:- 19045
No is Odd.

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DATE :
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→ Assignment → Q11:- Write a program to check if given no. is -ve, 0 or +ve.

→ Code →

```
#include <stdio.h>
int main()
{ int n;
  printf("Enter a No.:- ");
  scanf("%d",&n);

 if (n<0)
  { printf("Given No is Negative.");}
 if (n==0)
  { printf("Given No. is Zero.");}
 if (n>0)
  { printf("Given No. is Positive.");}

 return 0;
}
```

→ Output → Enter a No.:- -546
Given No. is Negative.

→ Assignment ⇒ 12 ⇒ Write a program to check if given char is a Vowel or Consonant.

→ Code ⇒

```
#include <stdio.h>
    int main()
    { char l;
      printf("Enter a letter:- ");
      scanf("%c", &l);

      if((l=='a')||(l=='e')||(l=='i')||(l=='o')||(l=='u')||
        (l=='A')||(l=='E')||(l=='I')||(l=='O')||(l=='U'))
      {printf("Letter is Vowel.");}

      else
      {printf("Letter is Consonant.");}

      return 0;
    }
```

→ Output:- Enter a letter:- U
Letter is Vowel.

22/9/26

→ Assignment ⇒ 13 ⇒ Write a program to print all prime nos. b/w 1 & N, where N is given by user.

→ Code ⇒

```
#include <stdio.h>
int main()
{ int e,n,i,x;
  printf("Enter the Value of N:- ");
  scanf("%d",&e)

  for(i=2; i<=e; i++)
  { n=1;
    for(x=2; x<i; x++)
    { if(i%x==0)
      { n=0;
        break;
      }
    }
    if(n==1)
    { printf("%d\n",i);
    }
  }
  return 0;
}
```

→ Output:- Enter a Value of N:- 10

2
3
5
7

→ Assignment ⇒ 14 ⇒ Write a program to find the factorial of a given no. using a loop.

→ Code ⇒

```
# include <stdio.h>
    int main()
      { int n, i;
        printf("Enter a No:- ");
        scanf("%d", &n);

        for (i=n; i>=1; i--)
        { n=n*i;
        }
        printf("%d", n)
        return 0;
      }
```

→ Output ⇒ Enter a No:- 5
120

→ Assignment ⇒ 15 ⇒ Write a program to generate Fibonacci Series (0,1,1,2,3,5,8...) upto N terms using loop.

→ Code ⇒

```
#include <stdio.h>
int main()
{ int s, e, i, x;
  printf("Enter ending (N) of Series :- ");
  scanf("%d", &e);

  s = 0;
  printf("%d", s);

  for (i = s+1; i <= e; i = x + s)
  { x = s;
    s = i;
    printf("%d", i);
  }
return 0;
}
```

→ Output ⇒ Enter ending (N) of Series :- 13
0 1 1 2 3 5 8 13

→ Assignment ⇒ 16 ⇒ Write a program to print table of given No.

→ Code ⇒

```
#include <stdio.h>
    int main()
    { int n, i, x;
      printf("Enter a No.:- ");
      scanf("%d", &n);

      for (i=1; i<=10; i++)
      { x = n*i;
        printf("%d ", x);
      }
      return 0;
    }
```

→ Output ⇒ Enter a No.:- 9

9
18
27
36
45
54
63
72
81
90

→ Assignment ⇒ 17 ⇒ Write a program to reverse a given Int. (eg:- 1234 → 4321). Check if it's a Palindrome.

→ Code ⇒

```
#include <stdio.h>
int main()
{   int n, i, x, a, b, c, d;
    printf("Enter a No.:- ");
    scanf("%d", &n);
    c = 0;
    a = n;
    d = n;

    x = n%10;
    b = x;

    for(i=1; i<=10; i++)
    {  n = n/10;
       c = c+1;
       if(n==0)
       {  break;
       }
    }
    for(i=1; i<=c; i++)
    {  x = (d/10);
       x = x%10;
       b = (b*10)+x;
       d = d/10;
    }
```

```
    b = b/10;
    printf("Reverse is :- %d \n", b);

    if (a == b)
    { printf("It is a Palindrome");
    }
    else
    { printf("It is not a Palindrome");
    }
    return 0;
}
```

→ Output-1 ⇒ Enter a No :- 121
Reverse is :- 121
It is a Palindrome.

→ Output-2 ⇒ Enter a No :- 123
Reverse is :- 321
It is not a Palindrome.

→ Assignment ⇒ 18 ⇒ Write a Program to check, if a given No. is an Armstrong No (0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 153, 370, 371 .....)

→ Code ⇒

```
#include <stdio.h>
#include <math.h>
    int main()
    { int n, a, b, i, x, c, j;
      printf("Enter a No:- ");
      scanf("%d", &n);
      c=0;
      x=n;
      for(i=1; i<=n; i++)
      { x=x/10;
        c = c+1;
        if(x==0)
        { break;
        }
      }
      a = n%10;
      b=n;
      j = pow(a, c);
      for(i=1; i<=c; i++)
      { b=b/10;
        a=b%10;
        j = j + pow(a,c);
      }
```

```
    if (n==0)
    { printf("This is an Armstrong No.");
    }
    else if (n==j)
    { printf("This is an Armstrong No.");
    }
    else
    { printf("This is not an Armstrong No.");
    }
     return 0;
}
```

→ Output ⇒ Enter a No :- 153
This is an Armstrong No.

→ Assignment ⇒ 19 ⇒ Write a program to display :-

```
*
**
***
****
*****
```

→ Code ⇒

```
#include<stdio.h>
int main()
{ int i,j;
  for(i=1; i<=5; i++)
  { for(j=1; j<=i; j++)
    { printf("*");
    }
    printf("\n");
  }
  return 0;
}
```

→ Output ⇒

```
*
**
***
****
*****
```

2019